

# ITA0612-Machine Learning

## CO1-AT1- APPLICATION BASED QUESTIONS

Submitted By : Shanuusree M (192412319)

Submitted To : Dr. Shri Vindhya A

Annexure A – Application-Based Questions

### QUESTIONS:

1. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?(5 Marks)
2. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)
  1. (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
  2. (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2 3. (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
  4. (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
  5. (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
  6. (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
  7. (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
  8. (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2
  9. i) Identify the epoch at which the model begins to overfit.
  10. ii) Suggest two effective techniques to reduce overfitting in this model.
  11. iii) Discuss how overfitting affects the model's generalization ability on unseen data.

3.Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

12. (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

13. (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63% 14. (c) Epoch 3:

Training Accuracy = 70%, Validation Accuracy = 68%

15. (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%

16. (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%

17. (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

18. (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

19. (h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%

20. i) Identify the epoch at which the model starts overfitting.

21. ii) Explain why validation accuracy decreases despite training accuracy increasing.

22. iii) Suggest two methods to improve validation performance.

ANSWERS:

# 1. Application of Bayes' Theorem in Machine Learning

**Given:**

Section selection probabilities:

- $P(A) = 1/4$
- $P(B) = 1/2$  (Section B is twice as likely to be selected) •  $P(C) = 1/4$

Probability of selecting a blue car from each section:

- $P(\text{Blue}|A) = 3/10$
- $P(\text{Blue}|B) = 6/15 = 2/5$
- $P(\text{Blue}|C) = 2/15$

**Using Bayes' Theorem:**

$$P(C|\text{Blue}) = [P(C) \times P(\text{Blue}|C)] / [P(A) \times P(\text{Blue}|A) + P(B) \times P(\text{Blue}|B) + P(C) \times P(\text{Blue}|C)]$$

Substitute the values:

$$P(C|\text{Blue}) = [(1/4 \times 2/15)] / [(1/4 \times 3/10) + (1/2 \times 2/5) + (1/4 \times 2/15)]$$

$$= (1/30) / (3/40 + 1/5 + 1/30)$$

Take the LCM = 120

$$= (1/30) / (9/120 + 24/120 + 4/120)$$

$$= (1/30) / (37/120)$$

$$= (1/30) \times (120/37)$$

$$= \mathbf{4/37}$$

$$= \mathbf{0.108 \approx 10.8\%}$$

**Answer:**

The probability that the selected blue car came from **Section C** is **4/37**

## 2. Training and Validation Loss Analysis

i) Identify the epoch at which the model begins to overfit.

The validation loss decreases until **Epoch 3 (2.0)** and starts increasing from **Epoch 4**, while the training loss continues to decrease.

**Answer:**

The model begins **overfitting at Epoch 4**.

ii) Suggest two effective techniques to reduce overfitting.

### 1. Early Stopping

- Stop training when the validation loss starts increasing.
- Prevents the model from learning noise in the training data.

### 2. Regularization (L2 or Dropout)

- L2 Regularization penalizes large weights.
- Dropout randomly deactivates neurons during training.
- Both techniques improve the model's generalization.

iii) Discuss how overfitting affects the model's generalization ability.

Overfitting occurs when the model learns the training data too closely, including noise and irrelevant patterns.

As a result:

- Training loss becomes very low.
- Validation loss increases.
- The model performs poorly on unseen or test data.
- Generalization ability decreases.

### 3. Training and Validation Accuracy Analysis

i) Identify the epoch at which the model starts overfitting.

Validation accuracy increases until **Epoch 5 (73%)** and then decreases, while training accuracy continues to improve.

**Answer:**

The model starts **overfitting from Epoch 6**.

ii) Explain why validation accuracy decreases despite training accuracy increasing.

The model begins memorizing the training data rather than learning meaningful patterns.

Therefore:

- Training accuracy continues to improve.
- Validation accuracy decreases because the model cannot generalize well to new, unseen data.

iii) Suggest two methods to improve validation performance.

#### 1. Early Stopping

- Stops training before the model starts overfitting.

#### 2. Data Augmentation

- Generates more diverse training samples through transformations such as rotation, flipping, cropping, or noise addition.
- Helps the model learn more generalized features